AI for Good Sandbox Hackathon – Team MAJLIS – MAJLIS: A Mandate-Gated Multi-Agent Crisis Council for Policy Gaps in AI-Driven Telecom Networks

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Github link: https://github.com/roshworldwide/majlis
Knowledge base: https://github.com/roshworldwide/majlis/tree/main/kb/export (chunks, manifest with URL/SHA-256/licence, datasheet)

Open the live simulation — no install: https://roshworldwide.github.io/majlis/

1. **Introduction**

   MAJLIS — "no mandate, no action." A council of AI agents (an operator's network copilot, five authorities modelled on UAE institutions, a Policy & Legal Adviser and a coordinator) handles a fictional AI-caused telecom crisis. No agent may act unless it cites the law that empowers it: the Y.3172 Policy node (P) — the ML-pipeline step where policies are applied to a model's output — checks every action, message and claim against a 4,919-chunk knowledge base. Whenever a mandate, channel or duty cannot be cited, MAJLIS logs a **typed policy gap** (one of the brief's six types) and retrieves how other jurisdictions closed it. Novelty: mandate-gated agents plus automatic, evidenced gap discovery with verified global examples, replayed under a reform pack to measure the effect.

What is real / what is simulated
Real: 100 ingested public documents with URL, SHA-256 and licence; every quote verified verbatim by the gate; cited mandates and channels; the hash-chained audit log; the code. **Simulated:** the scenario (fictional), all events, timings and KPIs (scenario parameters), the agents (simulations *modelled on* public mandates; they do not speak for any institution) and every metric (simulation outputs, not predictions).

2. **Description of the use case**

   AI-driven RAN energy saving in a critical mobile network during a mass event. On a fictional New Year's Eve in Downtown Dubai, an operator's ML energy-saving model (v2.3, deployed 48 hours earlier) misreads crowd-peak traffic and puts radio cells to sleep exactly when demand peaks. The fault crosses domains: telecom (call setup collapses) → emergency (emergency calls fail) → health (ambulance response slows) → public trust (cyber-attack rumours). Information arrives in nine waves, unevenly: the operator sees KPIs, the ambulance service sees missing calls, the cyber authority sees rumours, and nobody sees everything.

3. **Use case requirements**

   R0 KB-grounded agents adhering to UAE instruments, mapped to global practice · **R1** NetOps Resilience Copilot finds the root cause from Y.3172 pipeline telemetry before customers report it · **R2** Policy & Legal Adviser queryable by any agent at any wave · **R3** institutional agents modelled on TDRA, NCEMA, a health/ambulance authority, the Cybersecurity Council and the UAE Data Office · **R4** MLFO coordinator routes messages only along cited legal channels · **R5** chunked, uneven waves plus a facilitator console · **R6** SHA-256 hash-chained audit log with structured reasoning traces · **R7** Y.3172 + Readiness alignment · **R8** reference-compatible ChromaDB KB with a loader into the ITU reference server.

4. **Mapping to ITU-T Y.3172**

Node	MAJLIS component	REQ-ML
SRC	UAE, global and ITU sources; scenario telemetry and injects	COR
C	collect.py (manifest, SHA-256), scenario loader	INT
PP	legal-structure chunking, page/article tags, BM25 + nomic-embed-text	SPEC
M	LLM agents A1–A6 (qwen2.5:3b), Adviser A7 (qwen2.5:7b) + hybrid retrieval	MNG
P	Mandate & Citation Gate G1–G5: blocks and raises gaps	—
D	router: legal channels, latencies, informal re-routing	DEP
SINK	gap register, IAPs, audit chain, dashboard, Experience	—
MLFO	Coordinator A8: waves, visibility, termination	MNG
ML Sandbox	the simulator; reform replay = policy extrapolation	—

5. **Mapping to the ITU AI Readiness Report 2.0**

Perspective (brief)	Metric (dimensions.json)	MAJLIS evidence
Open data / open source	d1_m13 Open-Source Models	MIT code; open KB export
GenAI content ecosystem	d2_m04 Fake Detection	rumour wave; rebuttal gap
Contextualisation	d4_m02 Local Datasets; d6_m02	UAE layer; EN/AR labels
Integration / cross-domain	d3_m01 Integrated Workflow	telecom → emergency → health
Human supervision	d6_m06 Safety Integration; d8_m06	P-node gate; IAPs; facilitator
Policy opportunity	d10_m01, d10_m03, d10_m08	sandbox; reform replay; gap briefs
Social inclusion	d11_m13 Social Inclusion Impact	public emergency access KPI
Infrastructure & energy	d13_m07 Uptime; d13_m06	restoration time; energy conflict

6. **Requirements mapping to local policies**

Requirement	UAE instrument (article)	Status
Regulator oversight of licensees	Fed. Decree-Law 3/2003 (telecom) Art. 12, 14(3)	covered
Operator → regulator incident report	TDRA IA Regulation §2.3 (cyber); QoS policy §8 (quarterly)	partial
Crisis coordination, information duty	Decree-Law 35/2024 Art. 3(12), 6(1)	partial

Requirement	UAE instrument (article)	Status
Personal-data breach notice; transfers	Decree-Law 45/2021 (PDPL) Art. 9, 22–23	covered
Purpose limitation (AI training data)	PDPL Art. 5(2) “similar or close” purpose	partial
Human oversight of AI	UAE AI Charter 2024, principle 6 (guiding framework)	partial
False news in crises	Fed. Decree-Law 34/2021 Art. 52	covered
AI-incident notice; ML update validation	—	not found*
Operator ↔ emergency-service channel	—	not found*

*not found in the ingested public corpus (as of 10 Nov 2026); absence queries and their top hits are logged with each gap.

7. **Evaluation scenario covering technical solution and policy**

   Eight agents, nine waves (T+0 to T+90 simulated minutes), per-agent visibility. Baseline = the current UAE corpus; reform = RF1 AI-incident flash notice, RF2 mass-event safe-mode, RF3 pre-deployment ML-sandbox validation with in-country rollback (design parameters, modelled on cited global clauses). **Technical:** the copilot identified the root cause at T+0 from telemetry (correlation of sleep decisions with call-setup failures plus the change log). **Policy, baseline:** the gate blocked 4 actions, re-routed 3 messages without a legal channel, flagged 1 conflict(s) and raised 9 gaps of all six types; service restored at T+70. **Reform:** the ambulance service learned the root cause at T+5 instead of T+35, the crisis authority at T+5 instead of T+35, service restored at T+5; the lead ambiguity remained, as expected. **Expected vs actual:** 9 of 9 outcomes written before the run held (docs/scenario.md).

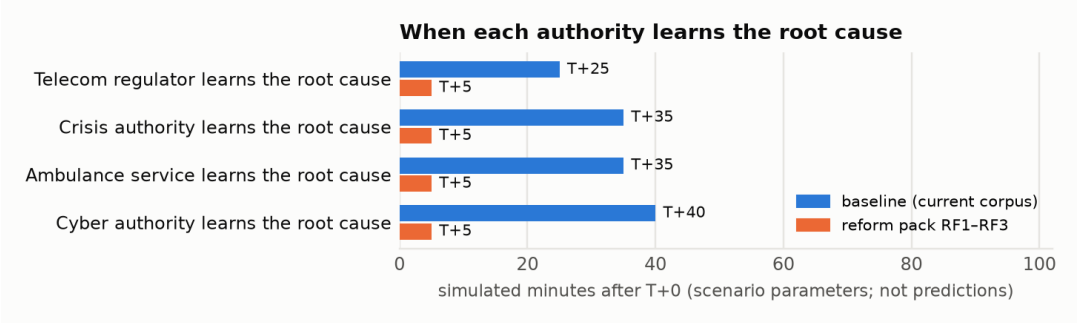


Figure 2 – Time to awareness, baseline vs reform pack (simulation outputs).

Agency: besides the 6 scripted questions, agents put 5 questions of their own to the Adviser; the facilitator can release waves, inject events and question the Adviser as any agent (majlis play / ask). **Auditability:** with the gate on, 7 unsupported claims (non-verbatim quotes or uncited “must/shall” statements) were caught and citation coverage was 0.931; with the gate off, 5 passed and 4 out-of-mandate actions executed. Three replays gave identical audit roots. **KB:** 61 gold questions, hybrid hit@5 0.951 (BM25 0.984, dense 0.918).

A1 asks the Adviser (W2): Is the operator required to notify any authority about an AI-caused network incident, and within what deadline? **A7:** According to [U3], the operator

is required to establish a Computer Security Incident Response Team (CSIRT) responsible for managing incidents. However, there is no explicit requirement in the UAE sources provided that mandates notification of any authority about an AI-caused network incident or specifies a deadline for such notifications. [verified sources: U3 UAE-TDRA-IAR-2020 ✓; G2 EU-AIACT-2024 Article 73 ✓]

A5 asks the Adviser (W3): Who has the mandate to publicly rebut false rumours of a cyberattack during a network outage? **A7:** [U3] Federal Decree-Law No. (34) of 2021 On Countering Rumors and Cybercrimes provides a framework for addressing false rumors but does not explicitly mandate who should publicly rebut them during a network outage. The Cybersecurity Authority [A5] at T+20 may have the authority to do so, given its role in cybersecurity, but this is not clearly stated. [verified sources: U3 UAE-CYBERCRIME-2021 Article 51, Article 52 ✓]

8. **Policy/standards gaps identified from the mapping exercise**

Type	Domain-level gap	Global example (verified clause)	IDs
legislative gap	No AI-incident notification duty or deadline for AI-driven network operations	NIS2 Art. 23: “without undue delay and in any event within 24 hours of becoming aware of the significant incident,…”	d10_m06 d10_m09
coordination problem	No channel between the network operator and emergency dispatch	47 CFR § 4.9(h): “shall provide a 911 outage notification to a potentially affected 911 special facility as soon as…”	d7_m01 d9_m08
unclear responsibility	Who leads when a telecom AI fault becomes a public emergency	NIS2 Art. 9(2): “it shall clearly indicate which of those authorities is to serve as the coordinator for the…”	d7_m01 d7_m05
conflicting objectives	Energy-efficiency goals vs network availability during mass events	UK Telecoms Security CoP v1.1 §2.70: “how easy it would be to revert back to a manual process”	d13_m07 d13_m06
standards gap	No mandated pre-deployment validation of ML model updates, and no standard AI-incident interface from the ML pipeline	ITU-T Y.3172 (ML sandbox): “An ML sandbox is an isolated domain which allows the hosting of separate ML pipelines to train,…”	d10_m01 d7_m04
technology gap	Vendor-held, offshore model and logs block forensics and rollback	AI Act Art. 12: “High-risk AI systems shall technically allow for the automatic recording of events (logs) over the…”	d13_m07 d10_m10
legislative gap	Unclear lawful basis for reusing subscriber location data to retrain network AI	AI Act Art. 10: “Training, validation and testing data sets shall be subject to data governance and management…”	d1_m10 d1_m09

Domain recommendations. (1) Introduce an event-driven AI-incident notification duty for AI systems that operate critical communications networks: a short early-warning deadline after awareness, a follow-up report and a final root-cause report, covering AI-

caused service-affecting incidents and not only cyber-attacks or personal-data breaches. (2) Establish a pre-agreed, two-way liaison between operators of AI-driven networks and emergency services: operators warn potentially affected emergency call centres within minutes of discovering an outage that may affect emergency-call delivery, emergency services can query network status, and uninterrupted emergency access is an explicit resilience objective. (3) Require pre-deployment validation of AI-driven ML model updates through a sandbox environment, as defined by ITU-T Y.3172, and mandate an AI-incident interface from the ML pipeline to authorities, similar to the architecture specified in ITU-T Y.3181.

Next steps. A draft ITU-T input proposes a standardised AI-incident notification interface from the ML pipeline (P node / MLFO) to national authorities as a Y.3172-family supplement (docs/standards_gap_contribution.md); a Kaleidoscope paper; and the 2027 multi-country exercise, where adding a country means adding a countries/<cc>/ folder.

References. Federal Decree-Law 3/2003 on telecommunications (consolidated); TDRA QoS Regulatory Policy (2019); UAE Information Assurance Regulation (TDRA); Federal Decree-Law 35/2024 (Civil Defence Authority, NCEMA); Federal Decree-Law 45/2021 (PDPL); Federal Decree-Law 34/2021; UAE AI Charter (2024); UAE Energy Strategy 2050; Regulation (EU) 2024/1689 (AI Act); Directive (EU) 2022/2555 (NIS2); Directive (EU) 2018/1972 (EECC); 47 CFR Part 4; UK Telecommunications Security Code of Practice v1.1; India Telecom Cyber Security Rules 2024; TRAI outage-reporting direction (2023); OECD AI Papers No. 34 (2025); ITU-T Y.3172; ITU-T Y.3173; ITU-T Y.3181. Full list with URLs, retrieval dates and SHA-256: kb/export/manifest.csv.